

design and PCBs. This software has been written by Jean Pierre Charra, a researcher at Laboratoire des Images et des Signaux and a teacher at IUT de Saint Martin d'Hères, France.

The software comprises a schematic editor (Eeschema), a board editor (pcbnew), Gerber viewer (Gerbview), a footprint selector (Cvpcb) and a project manager. The software supports multiple platforms and has been released under the GPL licence.

What's special?

The schematic editor of Kicad gives the option to create simple or hierarchical sheets. It allows to check circuits with an electrical rule check tool (ERC) and also to create netlists for pcbnew or for Spice. Eeschema helps to manage access to component documentation.

The board editor gives the option of 1 to 16 copper layers and an additional 12 technical layers. The files that are necessary to develop PCBs, such as Gerber files for photo-plotters, can be created. Using pcbnew, 3D views of PCBs can be created with its components. This 3D view is achieved by OpenGL.

The software contains a library manager and an editor for Eeschema and pcbnew. It has the option to edit the library, create a new library or delete the library items. The software also comes with documentation to understand the software.

Users' feedback

This program is suitable for the board editor. The user interface is very professional and has a schematic program bundled with a layout editor. The attributes in Eeschema are not easily moveable and are all accessible for editing. This feature does not exist in other open source software such as gEDA.

Another great feature of Kicad is the netlist. The netlist of SPICE seems to work fine with this software. It helps to create a netlist from within a program. The board editor of the software has a good user interface, which supports up to 16 layers. The DRC check of the software works fine both in real-time as well as when invoked

Checkout EFY-CD Page

Some PCB designing software highlighted in the 'Designing and Simulation' section of EFY are:

- Target 3001: May 2009
- Kicad: June 2009
- TinyCAD: June 2009
- Vutrac: July 2009
- CADSTAR 11.0: August 2009
- Sprint Layout 5.0: September 2009
- Dip Trace: October 2009

Check out the EFY-CD page in this issue also for some more software.

at the end of the layout.

Areas for improvement

The Eeschema object selection is a bit strange. It cannot be selected by clicking on the object, rather it has to be used by drag-and-drop option. There are also no keyboard short-cuts. The pins in Kicad are rigidly defined and new pin attributes cannot be created. Another problem with Kicad is that the symbol and footprint creation is not easily scriptable as with gEDA/gafPCB.

Kicad's symbol and footprint libraries hold multiple symbols together instead of holding each symbol/footprint in an individual file. The software does not compile as per the GNU build system, which can be a problem for the newbies. Undo features also need to be added in the software.

Home page: http://www.lis.inpg.fr/realise_au_lis/kicad/

Download URL: <ftp://iut-tice.ujf-grenoble.fr/caof>

Platform: Linux, Windows and Mac OS

PADS

PADS is from Mentor Graphics, and its latest version is 9.0. It gives the option of either entry-level PCB development or complex industry-level designs. The typical suite of PADS incorporates a HyperLynx pre- and post-layout signal integrity simulation, thermal analysis and analogue simulation, powerful autorouting features, unlimited database and layers, design reuse, advance manufacturing rules, 3D view, etc. The software is cost-effective as it can be customised.

What's special?

PADS is easy to use and provides a broad range of PCB development functionality such as schematic entry, component management, FPGA to PCB optimisation, analogue design, signal and power integrity analysis, PCB layout, manufacturing preparation and 3D visualisation.

PADS helps to layout and design complex circuit boards with high-speed applications and RF circuits. It also offers advanced design rules with real-time design rule checking, bi-directional cross probing, RF design functionality, split-plane generation, auto-dimensioning, direct DXF import into both the board and part library editors, advanced fabrication verification tool and 3D verification.

PADS has thermal analysis features, which help to analyse the thermal problems on all PCB layout environments, thus helping to resolve board and component overheating in the design process. The software also gives an option to include FPGA, schematic and PCB designs in a single tool.

Users' feedback

With the new PADS high-speed, autorouting feature, the design-cycle time can be reduced from two months to 40 hours. Companies usually prefer this software as it is the best for training their employees. It is also useful for newbies as it reduces design time by 40 per cent and can be easily customised.

Areas for improvement

By incorporating the HyperLynx signal integrity, PADS software can reduce the design spins and help make better upfront decisions about design strategies. It is also more expensive compared to similar software such as EAGLE.

Home page: <http://www.mentor.com>

Download URL: <http://www.mentor.com/products/pcb-system-design/design-flows/pads/pads-evaluation>

Platform supported: Windows ●

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